

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP30022
Subject Title	Scientific Instrumentation and Automation
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: AP20007
Objectives	a) Provide students with an understanding of the theory, the characteristics, the operation and the application of scientific instruments in measurement b) Introduce the techniques in analogue signal processing c) Describe the technical details of both the hardware and software in a computer interfacing system d) Apply knowledge in Lab View and instrumentation to the development of laboratory automation e) Provide examples of the applications of AI in the scientific instrumentation and automation
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Describe and explain the theory, characteristics, operation and applications of the basic instruments in electrical measurements; b) Solve problems in analogue instrumentation using Golden rules, Ohm's law and Kirchhoff's rules for circuits involving operational amplifiers, capacitors and resistors; c) Describe the technical details of both the hardware and software in a computer interfacing system; d) Apply knowledge in Lab View and instrumentation to the development of laboratory automation; e) Design simple interfacing systems; f) understand the applications of AI in the scientific instrumentation and automation

Subject Synopsis/ Indicative Syllabus	Analog circuit: I-V characteristics of general nonlinear components (diode and transistor); rectifier circuits; clipping and clamping circuits; operational amplifiers – frequency response characteristics, linear and nonlinear circuits, active filters and oscillators, signal generators. Measurement techniques and electronic instruments: Measurement and error; displacement and thermal transducers and their applications. LabVIEW: graphical programming, virtual instrumentation, data acquisition, instrument control. Interfacing techniques for engineering/physics applications: input and output devices and their interfacing. AI applications in the scientific instrumentation and automation																																						
Teaching/Learning Methodology	Lecture: Background knowledge behind all experiments will be systematically introduced in lectures. Class work and assignments related to the content of lectures will be used to enhance students learning. Lab session: Experiments are essential for students to relate the concepts to practical applications and they are exposed to hands-on experience and proper use of equipment and also analytical skills on interpreting experimental results.																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Continuous assessment</td><td>50</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Practical test</td><td>50</td><td>√</td><td>√</td><td>√</td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessment includes assignments and laboratory reports. Assignments will strengthen the students’ basic knowledge and the analytical skills to solve the problems related to this subject. Lab session is useful to assess students’ experimental skills and knowledge learned from the lectures and laboratory works, which is mainly the design and control of interfaces between computer and scientific instruments. Practical Test will review their understanding of the course and assess their ability to solve problems, in particular using graphical programming. Hence, the proposed assessment methods are necessary to	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Continuous assessment	50	√	√	√	√	√	√	2. Practical test	50	√	√	√				Total	100						
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																																			
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2. Practical test	50	√	√	√																																			
Total	100																																						

	assess the intended learning outcomes.	
Student Study Effort Expected	Class contact:	
	▪ Lecture & Tutorial	30 Hrs.
	▪ Lab	18 Hrs.
	Other student study effort:	
	▪ Self-study	72 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	Operational Amplifiers & Linear Integrated Circuits: Theory and Application / 3E, Fiore, James A.H. Robbins and W.C. Miller, "Circuit Analysis: Theory and Practice", 2nd Edition, Thomson Learning, 2000. "The LabVIEW Style Book" by Peter A. Blume. (Prentice Hall, 2007). "LabView for Everyone: Graphical Programming Made Easy and Fun" by Jeffrey Travis and Jim Kring (Prentice Hall, 2006).	